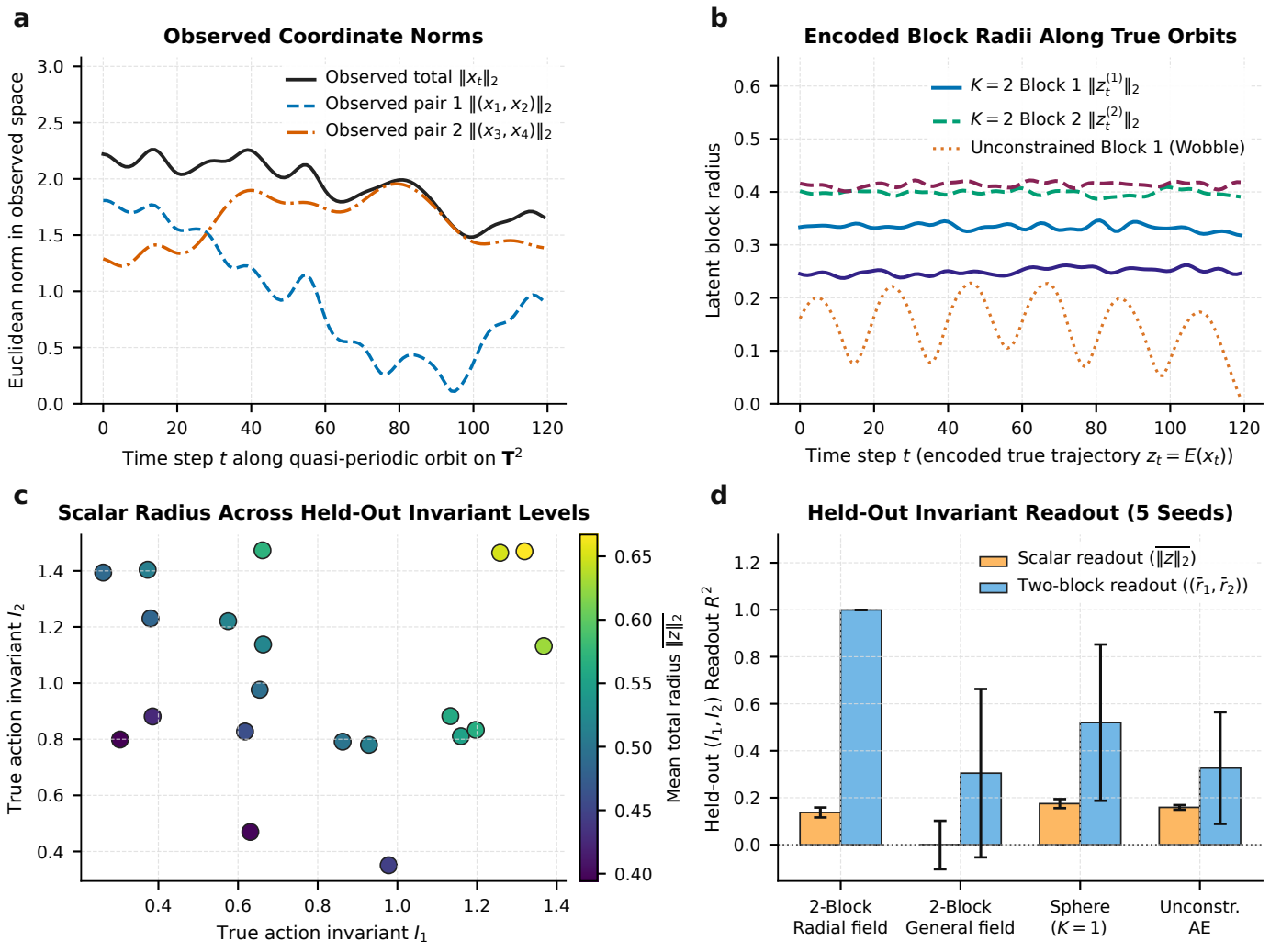
a
Observed Coordinate Norms
Observed total $\|x_t\|_2$
Observed pair 1 $\|(x_1, x_2)\|_2$
Observed pair 2 $\|(x_3, x_4)\|_2$
Euclidean norm in observed space
Time step t along quasi-periodic orbit on $\mathbf{T}^2$
b
Encoded Block Radii Along True Orbits
$K = 2$ Block 1 $\|z_t^{(1)}\|_2$
$K = 2$ Block 2 $\|z_t^{(2)}\|_2$
Unconstrained Block 1 (Wobble)
Latent block radius
Time step t (encoded true trajectory $z_t = E(x_t)$)
c
Scalar Radius Across Held-Out Invariant Levels
True action invariant I_2
True action invariant I_1
Mean total radius $\overline{\|z\|_2}$
d
Held-Out Invariant Readout (5 Seeds)
Scalar readout ($\overline{\|z\|_2}$)
Two-block readout ($(\bar{r}_1, \bar{r}_2)$)
Held-out (I_1, I_2) Readout R^2
2-Block Radial field
2-Block General field
Sphere ($K = 1$)
Unconstr. AE